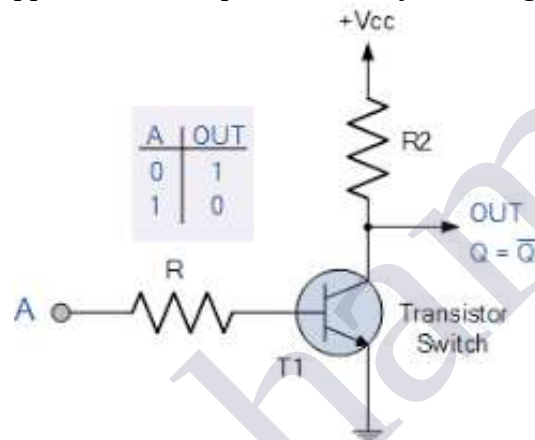


1.

a) Explain how transistor is used as inverter? Draw the circuit of a CMOS switch.

A transistor acts as an inverter by flipping the input signal. When we connect an input signal to the base of an NPN transistor through a resistor, and take the output from the collector, the behavior is straightforward. When the input is high (1), the transistor turns on, allowing current to flow from the collector to the emitter, which pulls the output to low (0). Conversely, when the input is low (0), the transistor turns off, stopping the current flow, and the output is pulled high (1) by the power supply through a resistor. This results in the output being the opposite of the input, effectively inverting the signal.



Circuit of a CMOS switch:

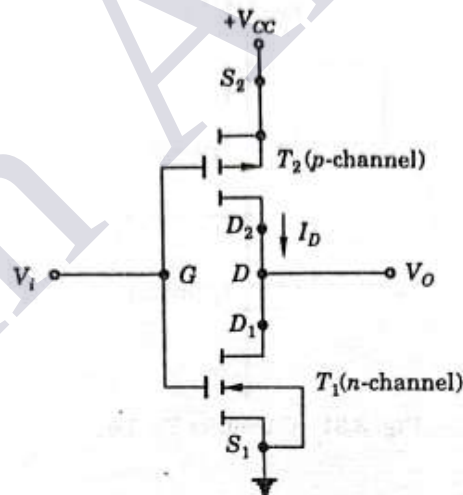


Fig. 3.33 A CMOS switch

b) Discuss on speed of operation, noise immunity and figure of merit in digital logic families.

4.2.1 Speed of Operation

The speed of a digital circuit is specified in terms of the propagation delay time. The input and output waveforms of a logic gate are shown in Fig. 4.1. The delay times are measured between the 50 per cent voltage levels of input and output waveforms. There are two delay times: t_{pHL} , when the output goes from the HIGH state to the LOW state and t_{pLH} , corresponding to the output making a transition from the LOW state to the HIGH state. The propagation delay time of the logic gate is taken as the average of these two delay times.

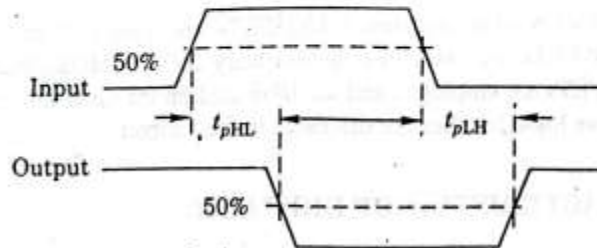


Fig. 4.1 Input and output voltage waveforms to define propagation delay times.

4.2.3 Figure of Merit

The figure of merit of a digital IC is defined as the product of speed and power. The speed is specified in terms of propagation delay time expressed in nanoseconds.

$$\text{Figure of merit} = \text{propagation delay time (ns)} \times \text{power (mW)}$$

It is specified in pico joules ($\text{ns} \times \text{mW} = \text{pJ}$)

A low value of speed-power product is desirable. In a digital circuit, if it is desired to have high speed, i.e. low propagation delay, then there is a corresponding increase in the power dissipation and vice-versa.

4.2.6 Noise Immunity

The input and output voltage levels defined above are shown in Fig. 4.3. (Stray electric and magnetic fields may induce unwanted voltages, known as *noise*, on the connecting wires between logic circuits. This may cause the voltage at the input to a logic circuit to drop below V_{IH} or rise above V_{IL} and may produce undesired operation.

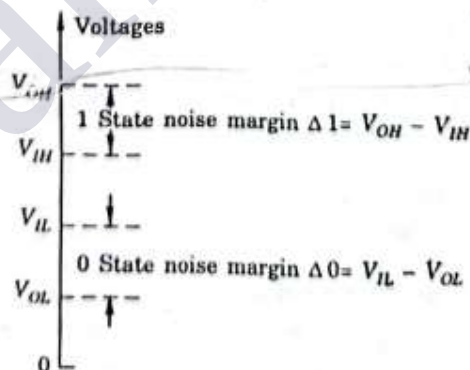
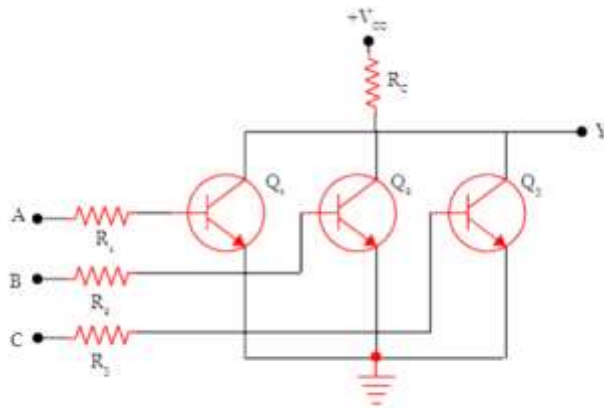


Fig. 4.3 Voltage levels and noise margins of ICs.

tion. The circuit's ability to tolerate noise signals is referred to as the *noise immunity*, a quantitative measure of which is called *noise margin*. Noise margins are illustrated in Fig. 4.3.

c) Draw the circuit diagram of 3 inputs RTL NOR gate and explain how it works.



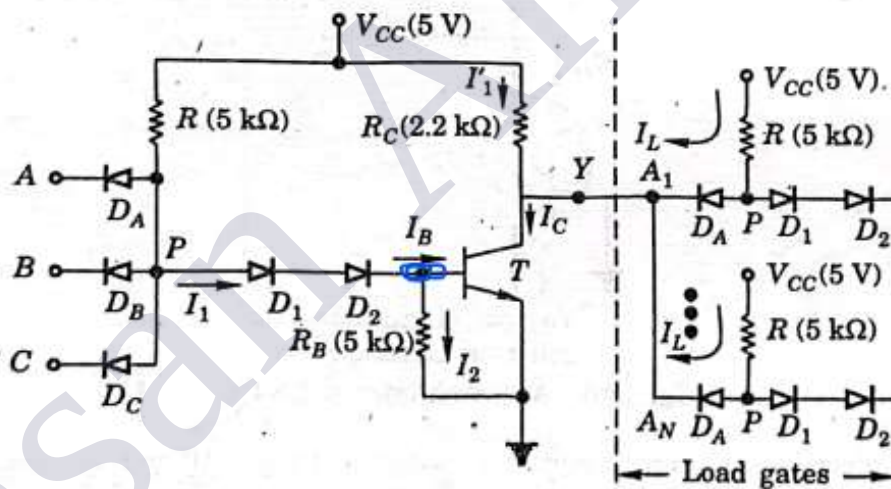
Input			Output
A	B	C	Y
0	0	0	1
1	0	0	0
0	1	0	0
1	1	0	0
0	0	1	0
1	0	1	0
0	1	1	0
1	1	1	0

2.

a) For the DTL NAND gate where diode and transistors parameters are given as Diode:

- ✓ Voltage across the diode=0.7V; Cut-in voltage=0.6V
- ✓ Transistor: Cut-in voltage=0.5V; $V_{BE, \text{sat}} = 0.8\text{V}$; $V_{CE, \text{sat}} = 0.2\text{V}$; $h_{FE} = 20$; $R_C = 2\text{ K}\Omega$; $R_B = 4\text{ K}\Omega$
- ✓ Resistor connected with $V_{CC} = 4\text{ K}\Omega$. Calculate i) fan-out ii) noise-margins iii) average power- P.

Redraw the circuit and modify the value for above value and proceed.



(i) If all the inputs are HIGH, the input diodes are reverse-biased. Assuming diodes D_1, D_2 to be conducting and T to be in saturation, the voltage $V_p = 0.7 + 0.7 + 0.8 = 2.2$ V.

Writing Kirchhoff's current law (KCL) equation at the base of T ,

$$I_B = I_1 - I_2$$

where

$$I_1 = \frac{V_{CC} - V_p}{R} = \frac{5 - 2.2}{5} = 0.56 \text{ mA}$$

and

$$I_2 = \frac{V_{BE, \text{sat}}}{R_B} = \frac{0.8}{5} = 0.16 \text{ mA}$$

which gives a base current $I_B = 0.4$ mA. The collector current (without load gates connected) is

$$I_C = \frac{V_{CC} - V_{CE, \text{sat}}}{R_C} = \frac{5 - 0.2}{2.2} = 2.182 \text{ mA}$$

Since $h_{FE} \cdot I_B = 30 \times 0.4 = 12$ mA is greater than I_C (2.182 mA), it is confirmed that the transistor is in saturation and the output is in LOW state. Now, if N load gates are fed from this gate, the input diodes of the driven gates will conduct through the output transistor T , i.e. T acts as a sink for the current in the input to the gates it drives. Assuming that all the other inputs to each of the load gates are HIGH

except the one driven by T , the current $I_L = \frac{V_{CC} - V_p}{R} = \frac{5 - 0.9}{5} = 0.82$ mA. This current is referred to as *standard load*. The fan-out is given by $I_C \leq h_{FE} I_B$, or $0.82 N + 2.182 \leq 12$ mA or $N < 12$ since N must be an integer. A conservative choice is $N = 10$. The Maximum collector current rating of T must be about 12 mA.

(ii) If at least one of the inputs is LOW, the corresponding input diode conducts and $V_p = 0.2 + 0.7 = 0.9$ V. The minimum voltage required for D_1, D_2 , and T to be conducting is $0.6 + 0.6 + 0.5 = 1.7$ V, which confirms that D_1, D_2 are nonconducting and hence T is cut-off. Consequently, the output voltage is V_{CC} (5 V) if the load gates are not connected.

If the load gates are connected, the input diodes of the load gates are nonconducting, which means the reverse-saturation current of these diodes must be supplied through the collector resistor R_C , which will produce a voltage drop across R_C and consequently the output voltage corresponding to HIGH state will be a little less than V_{CC} . The maximum current which can be supplied by the gate will depend upon V_{OH} . The fan-out is determined on the basis of maximum current.

(b) (i) If all the inputs are HIGH, the output is LOW. Since $V_p = 2.2$ V, the input diodes are reverse-biased by $5 - 2.2 = 2.8$ V. Since the cut-in voltage of the diode is 0.6 V, a negative noise spike of at least 3.4 V present at the input will cause malfunction of the circuit, i.e. the 0 level noise margin $\Delta 0 = 3.4$ V.

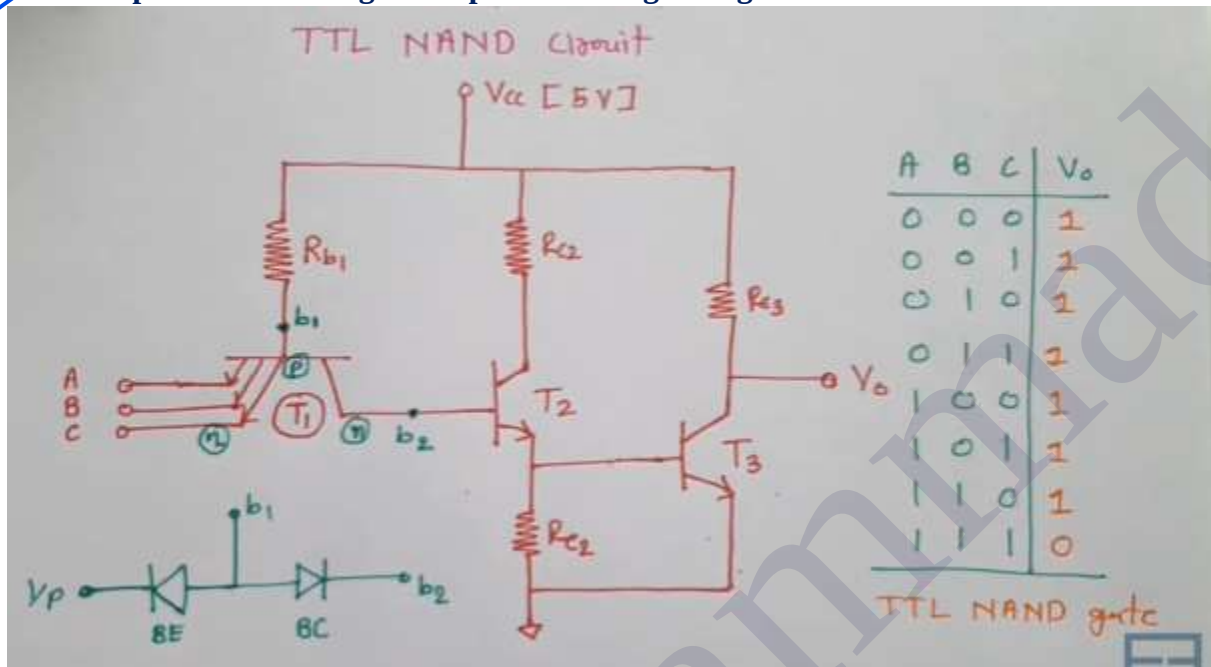
(ii) If at least one input is LOW, the output is HIGH. Since $V_p = 0.9$ V and a voltage of at least 1.7 V [part a (ii)] is required for D_1, D_2 , and T to conduct, therefore a positive noise spike of at least 0.8 V will cause malfunction of the circuit, i.e. the 1 level noise margin $\Delta 1 = 0.8$ V.

(c) The power $P(0)$ when the output is LOW is given by $P(0) = V_{CC} (I_1 + I'_1) = 5 (0.56 + 2.182) = 13.71$ mW. When the output is in the HIGH state at least one of the input diodes conduct. Therefore, $I_L = 0.82$ mA and $I'_1 = 0$. Hence $P(1) = (0.82)(5) = 4.1$ mW.

If we assume that the occurrence of LOW and HIGH is equally likely then the average power is

$$P_{av} = \frac{P(0) + P(1)}{2} = \frac{13.71 + 4.1}{2} = 8.905 \text{ mW}$$

a) How 3-input TTL NAND gate is operated in digital logic families?



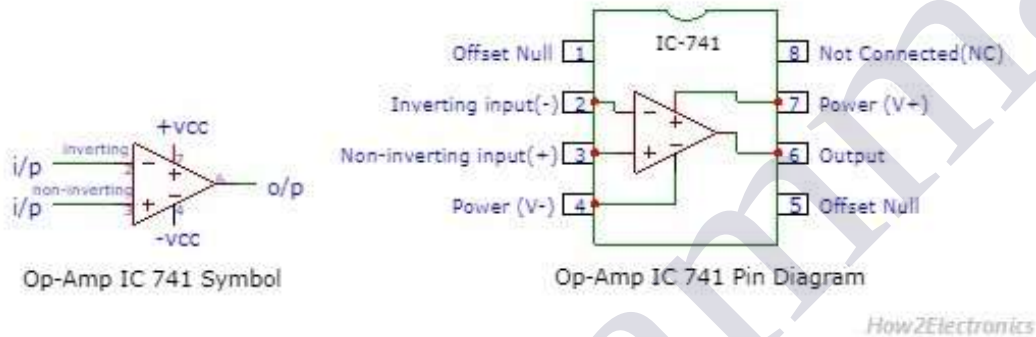
b) Discuss about the saturated bipolar logic families.

Saturated bipolar logic families use transistors that are fully turned "on" (saturated) to perform logic operations. Transistor-Transistor Logic (TTL) is one such family, using two transistors in each logic gate. It is fast and commonly used but consumes more power because the transistors continuously draw current. Emitter-Coupled Logic (ECL), another example, avoids saturation for even faster operation. ECL is very quick and used in high-speed applications, but it consumes a lot of power. In summary, these logic families leverage saturated transistors for speed, with TTL being widely used and ECL designed for high-performance needs.

3.

a) Show the pin configuration of a typical op-amp with its function.

The symbol and pin configuration of op-amp 741 is shown below. The diagram comprises eight pins. Among these, pins 2, 3, and 6 are the most significant pins, where pins 2 and 3 represent inverting and non-inverting terminals respectively, and pin 6 represents output voltage. Pin 8 is inactive in the circuit. The number 741 in the name indicates that there are 7 active pins, 4 pins (pin 2, 3, 4, 7) are capable of taking input and 1 pin (pin 6) is an output pin. The triangular form in the IC represents an op-amp integrated circuit.

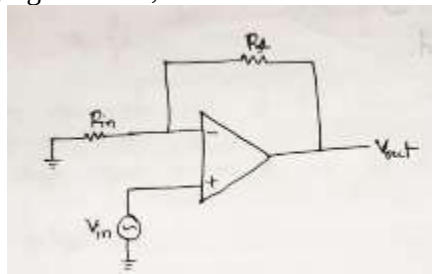


Functionality of the pin diagram:

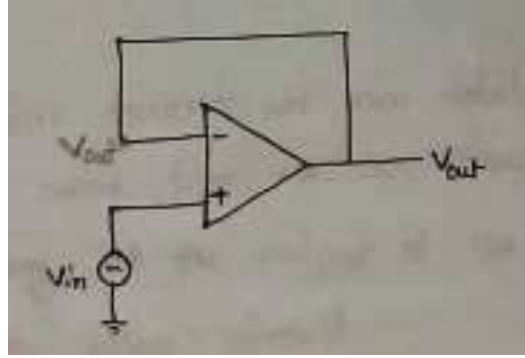
- **Pin 1 and 5 (Offset Null):** Used to adjust the offset voltage.
- **Pin 2 (Inverting Input):** Receives the input signal to be inverted.
- **Pin 3 (Non-Inverting Input):** Receives the input signal to be amplified without inversion.
- **Pin 4 (V- or Negative Power Supply):** Connects to the negative power supply.
- **Pin 6 (Output):** Gives the amplified output signal.
- **Pin 7 (V+ or Positive Power Supply):** Connects to the positive power supply.
- **Pin 8 (NC - Not Connected):** Not used in the circuit

b) Draw the circuit of an Op-Amp as voltage follower and find an expression for its voltage gain.

The non-inverting op-amp configuration is,



This operational amplifier can be used to design a voltage follower. If we make the $R_f=0$ and $R_{in}=\infty$ then the circuit becomes like this:



Here, we have shorted this output terminal with the inverting terminal. So, at the inverting terminal the voltage become V_{out} . Because of the feedback, we have virtual short between inverting and non-inverting terminal. So we can say that,

$$V^- = V^+ \\ V^- = V_{in} = V_{out}$$

It means, whatever voltage is applied to the non-inverting terminal, it will appear at the output voltage. So, we can say that, the output voltage follows the input voltage and that is why this circuit is called as the voltage follower circuit.

The voltage gain from the first circuit is $V_{OL} = -(R_f/R_{in})$

- c) Draw the circuit for a non-inverting amplifier and explain its operations.

An amplifier which amplifies the input without producing any phase shift between input and output is called **Noninverting Amplifier**.

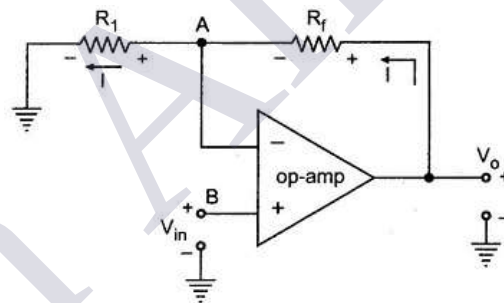


Fig. 2.20 Noninverting amplifier

The positive sign indicates that there is no phase shift between input and output. The input and output waveforms are shown in the Fig. 2.21.

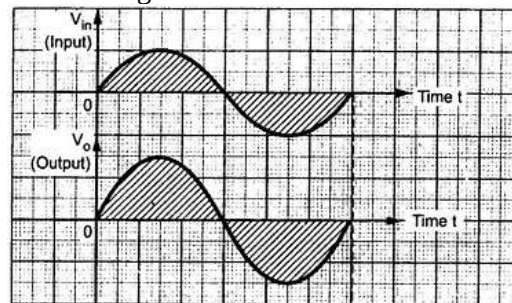


Fig. 2.21

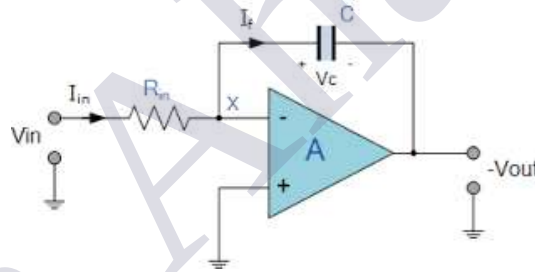
4.

a) Show the differences between open loop gain and close loop gain.

S. No	Open Loop Systems	Closed Loop System
1.	These are not Reliable.	These are Reliable.
2.	It is easier to build.	It is difficult to build.
3.	If calibration is good, they perform accurately.	They are accurate because of feedback.
4.	Open loop systems are generally more stable.	These are less stable.
5.	Optimization is not possible.	Optimization is possible.

b) Draw the circuit diagram of integrator and differentiator and explain how it works.

Op-amp Integrator: if we replace the feedback resistor of the inverting configuration with a capacitor, then we get the integrator circuit.



Let us consider the current flowing through R is I_{in} and current flowing through the capacitor is I_c . The capacitor C_F is connected as the feedback. Because of the feedback, the node x is virtually grounded. So, at this node the potential is 0.

Applying KCL to the node,

$$\begin{aligned}
 I_{in} - I_c &= 0 \\
 \Rightarrow I_{in} &= I_c \\
 \Rightarrow \frac{V_{in} - 0}{R} &= I_c
 \end{aligned}$$

We know capacitor current can be given by,

$$I_c = C \frac{dV_c}{dt} \quad \text{where, } V_c \text{ is the voltage at capacitor}$$

Putting this value,

$$\begin{aligned}
 \frac{V_{in}}{R} &= C_F \times \frac{dV_c}{dt} \\
 \Rightarrow \frac{V_{in}}{R} &= C_F \times \frac{d}{dt} [0 - V_{out}] \\
 &= -C_F \times \frac{dV_{out}}{dt}
 \end{aligned}$$

charge across the capacitor is,

$$Q = CV$$

$$\Rightarrow \frac{dQ}{dt} = C \frac{dV_c}{dt} = I_c \quad [\text{taking derivative}]$$

rate of change of charge

$$\frac{dV_{out}}{dt} = -\frac{1}{R_F} \times V_{in}(t)$$

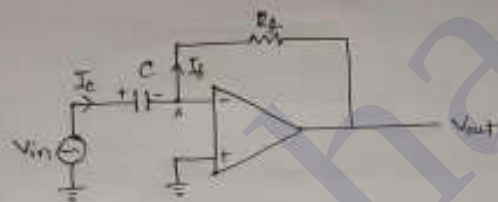
Taking integration for both sides,

$$V_{out}(t) = -\frac{1}{R_F} \int V_{in}(t) dt$$

Here by this equation, we get the integration of input voltage times the factor $\frac{1}{R_F}$ as the output voltage.

c) Op-Amp differentiator:

If we interchange the position of the capacitor and resistor of the integrator circuit, we get a simple differentiator circuit.



The current flowing through the capacitor is I_C and through R_F is I_F . We assume that this is an ideal op-amp, so no current is flowing in this op-amp. Due to the negative feedback the node x is virtually grounded and its potential is 0.

Applying KCL at x ,

$$I_C - I_F = 0$$

$$\Rightarrow I_C = I_F$$

$$\Rightarrow C \frac{dV_C}{dt} = I_F \frac{0 - V_{out}}{R_F}$$

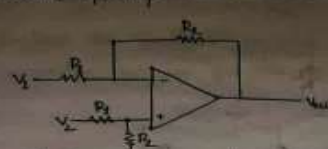
$$\Rightarrow C \frac{dV_{in}}{dt} = -\frac{V_{out}}{R_F}$$

$$\Rightarrow V_{out} = -R_F \times C \times \frac{dV_{in}}{dt}$$

So, in this circuit, the output voltage is the differentiation of the input voltage times $R_F C$ factor.

d) Draw the circuit diagram of subtractor and explain how it works.

Subtractor Op-Amp



This circuit is the combination of inverting and non-inverting configuration. Applying superposition theorem for V_{out} , let us consider $V_2 = 0$ and V_1 is acting alone. In this configuration, the output voltage is V_{o1} .

$$V_{o1} = \left(-\frac{R_2}{R_1}\right) V_1 \quad [\text{as it is the inverting configuration}]$$

Similarly, considering $V_1 = 0$ and V_2 is acting alone, then the output voltage of this configuration V_{o2} is,

$$V_{o2} = \left(1 + \frac{R_2}{R_1}\right) V_2$$

So, if these are acting simultaneously, then

$$\begin{aligned} V_{out} &= V_{o1} + V_{o2} \\ &= \left(-\frac{R_2}{R_1}\right) V_1 + \left(1 + \frac{R_2}{R_1}\right) V_2 \end{aligned}$$

Here we can see that, the gain for both cases are different. If we want the gain to be same for both inputs, and if we want the V_{out} as $K(V_2 - V_1)$ i.e. the difference between two input voltages times K , then we need to make sure the gain of the two input voltages are equal. So, in order to make the term $\left(1 + \frac{R_2}{R_1}\right)$ equal to $\frac{R_2}{R_1}$,

$$\begin{aligned} 1 + \frac{R_2}{R_1} &= \frac{R_2}{R_1} \\ &= \frac{R_1 + R_2}{R_1} \\ &= \frac{R_1 + R_2}{R_1} \times \frac{R_1}{R_1 + R_2} = \frac{R_2}{R_1} \end{aligned}$$

Here, we have applied the voltage divider instead of applying the voltage source V_2 directly. So, Now

$$\begin{aligned} V_{out} &= \left(-\frac{R_2}{R_1}\right) V_1 + \left(\frac{R_2}{R_1}\right) V_2 \\ \Rightarrow V_{out} &= \frac{R_2}{R_1} (V_2 - V_1) \end{aligned}$$

So, here the output voltage is opera the difference between the two voltage multiplied by the factor of $\frac{R_2}{R_1}$.

5.

a) Find the analog output voltage of a 3-bit D/A converter for all possible inputs. Assume $K=1$.

Answer is partially matched with 5(b). In 5(b) we deduce a general equation. Following this equation we make $K=1$ and $N = 3$ then we find output voltage for each combination.

For a 3-bit DAC, there are $2^3 = 8$ possible input combinations. Let's represent the digital input as b_2, b_1, b_0 , where b_2 is the most significant bit and b_0 is the least significant bit.

The analog output voltage can be calculated using the following formula:

$$V_{out} = K \times (b_2 \times 2^2 + b_1 \times 2^1 + b_0 \times 2^0)$$

Since $K=1$, the formula simplifies to:

$$V_{out} = b_2 \times 4 + b_1 \times 2 + b_0$$

Now, let's calculate the analog output voltage for all possible input combinations:

b_2	b_1	b_0	V_{out}
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

b) Design a weighted-resistor D/A converter and explain how it works.

The circuit of Fig. 10.1 can be used for converting the digital input to analog output which operates according to the above principle. This circuit is referred to as a *weighted-resistor D/A converter* since the resistance values are weighted in accordance with the binary weights.

In the circuit Fig. 10.1, the digital inputs (1 or 0) operate the switches. A switch is thrown to position 1 or 0 for a digital input corresponding to that bit being 1 or 0.

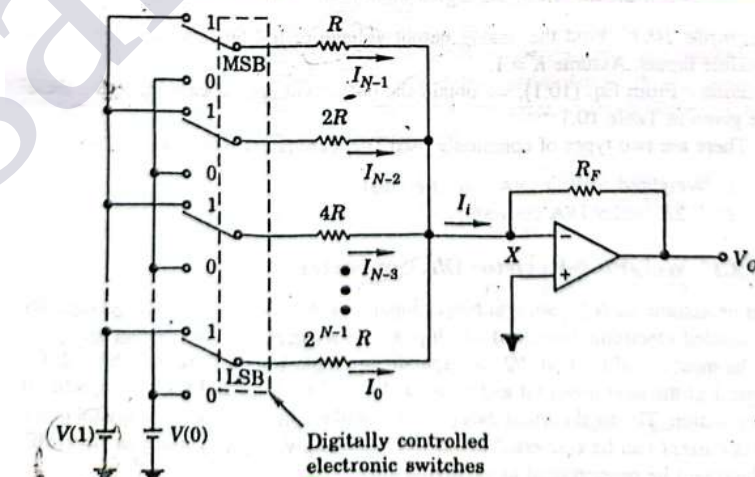


Fig. 10.1 Weighted-resistor D/A converter.

respectively. The voltage applied to a resistor is $V(1)$ if the switch connected to it is in position 1 and $V(0)$ if it is in position 0. The current, I_i is given by

$$I_i = I_{N-1} + I_{N-2} + I_{N-3} + \dots + I_2 + I_1 + I_0 \quad (10.2)$$

where

$$\left. \begin{aligned} I_{N-1} &= V_{N-1}/R \\ I_{N-2} &= V_{N-2}/2R \\ I_{N-3} &= V_{N-3}/2^2 R \\ &\vdots \\ I_0 &= V_0/2^{N-1} R \end{aligned} \right\} \text{where } V_n = V(1) \text{ if } b_n = 1 \\ = V(0) \text{ if } b_n = 0 \quad (10.3)$$

For straight binary input, $V(0) = 0$ and $V(1) = -V_R$, and the output voltage V_O is given by

$$V_O = -(-V_R) \left(\frac{R_F}{R} b_{N-1} + \frac{R_F}{2R} b_{N-2} + \frac{R_F}{2^2 R} b_{N-3} + \dots + \frac{R_F}{2^{N-1} R} b_0 \right) \quad (10.4)$$

which is of the same form as Eq. (10.1)

$$\text{with} \quad K = \frac{R_F}{2^{N-1} R} \cdot V_R \quad (10.5)$$

The output swings in only one direction and therefore is *unipolar*. If it is required to convert digital data in bipolar format such as in sign-magnitude, 1's complement or 2's complement format, then $V(0) \neq 0$. In such cases, $V(0)$ is used to *offset* the output swing.

With $V(1)$ and $V(0)$ as the voltages applied to the resistor network for 1 and 0, respectively, the output voltage V_O of Fig. 10.1 is given by

c) How offset voltage is used in D/A converter?

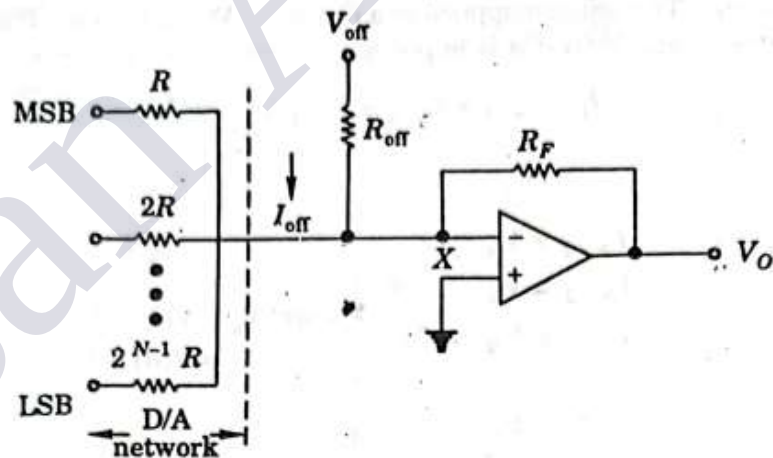


Fig. 10.2 Circuit used to offset the output voltage.

d) What is quantization error?

Quantization error happens when an analog signal is converted into a digital signal in an Analog-to-Digital (A/D) converter. It occurs because the A/D converter can only represent the signal in discrete steps, so small differences between the actual signal and its digital version are lost. This difference between the actual analog value and the closest digital value is the quantization error.

Example:

Imagine you are measuring temperature with a thermometer that can only show whole numbers (e.g., 25°C, 26°C). If the actual temperature is 25.5°C, you would round it to either 25°C or 26°C. The difference between the actual 25.5°C and the rounded number is the quantization error.

How to reduce quantization error:

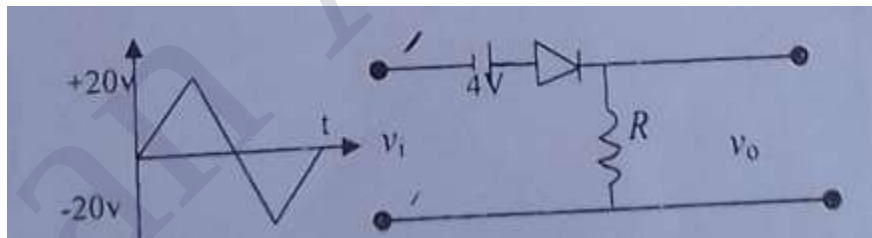
1. **Increase resolution:** By using more bits in the A/D converter, the number of steps increases, making each step smaller. This reduces the gap between the actual signal and the digital value.

For example, a 3-bit converter can represent 8 levels, while a 4-bit converter can represent 16 levels. More levels mean smaller errors.

Signal conditioning: Amplifying or filtering the signal before conversion helps by ensuring that the signal is well-matched to the converter's input range, reducing errors

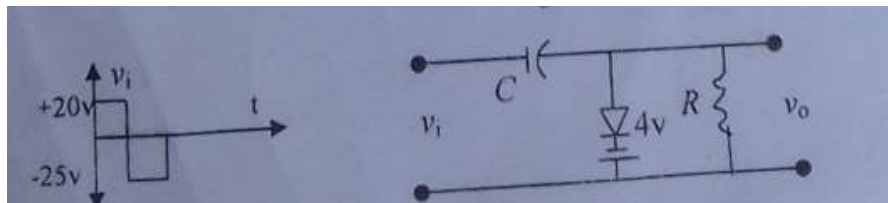
6.

- a) Analyze the clipper network with given input and determine the output waveform of the given network.



The positive half cycle result is $V_i + 4 = 24$ for negative half cycle the $V_o = V_i$ and the transition voltage is $V_i = -4$. Answer needed to recheck.

- b) Analyze output voltage for given Clamping network where $f=2500\text{Hz}$, $R=25\text{ k}\Omega$, $C=10\text{nF}$?



$5T = 5 \times R \times C = 5 \times 25 \times 1000 \times 10 \times 10^{-9} = 1.25 \times 10^{-3}$ ['T' is the symbolic notation of Tau]. Positive output voltage is -4 v and negative output voltage is -49.

c) How resistors and capacitors are considered for clippers.

In a clipper circuit, resistors and capacitors work together to shift the voltage level of a signal.

- ✓ **Resistor:** Controls how quickly the capacitor charges or discharges. It's like a gatekeeper for the flow of electric current.
- ✓ **Capacitor:** Stores and releases electrical energy. It decides when the voltage level shifts.

Example: Imagine we have a signal that swings between -5V and +5V. We want to move this signal up so that it swings between 0V and +10V instead.

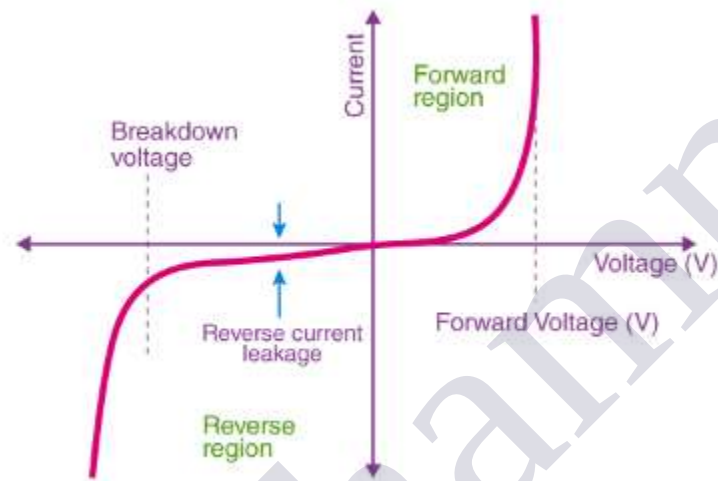
- Capacitor stores energy from the signal during the negative swing (-5V).
- Resistor controls how fast this stored energy is released when the signal moves positive.

Together, they "clip" the signal so that it starts from 0V instead of going into negative values.

1.

a) Discuss the switching characteristics of a semiconductor diode.

The V-I characteristics curve of a semiconductor diode is given below. This characteristic curve is a typical explanation for the V-I characteristic of a semiconductor diode. Current in the semiconductor diode starts to conduct when the current exceeds the threshold of the forward voltage, which is mentioned by the manufacturer.

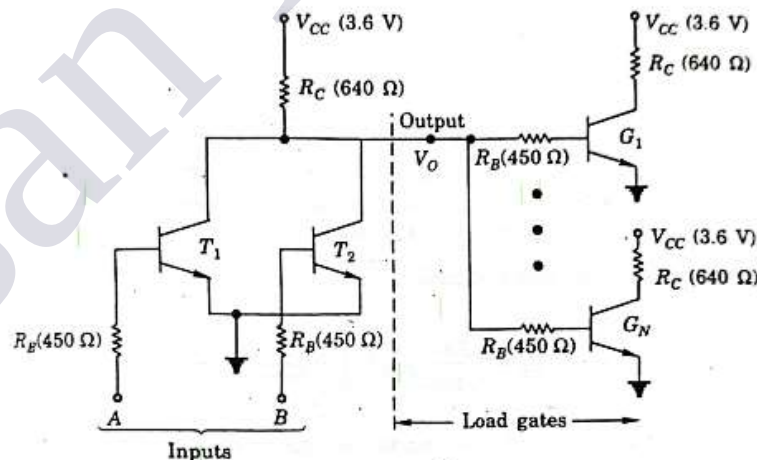


b) Draw the circuit diagram of 3-input RTL NOR Gate. Write down the working principle of the 3-input RTL NOR Gate.

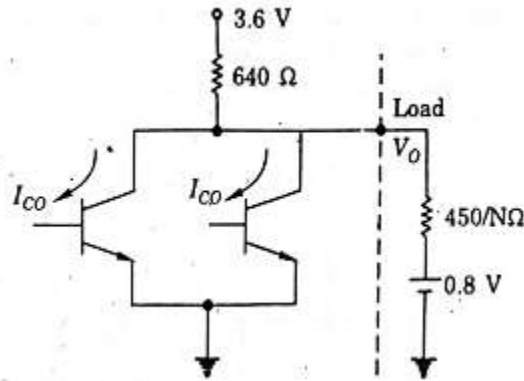
Already noted.

c) Calculate the noise margin of the RTL NOR gate if the parameters are considered as $V_{CC} = 5.0$; $R_L = 360 \text{ Ohm}$; $R_B = 450 \text{ Ohm}$; Cut in voltage of transistor $= 0.5 \text{ V}$; $N=5$; $h_{FE} = 10$.

Follow the format and adjust the value with this solution and redraw the circuit.



After simplification we find the simplified circuit like this:



The base current for each load transistor is

$$I_B = \left(\frac{3.6 - 0.8}{640 + \frac{450}{N}} \right) \cdot \left(\frac{1}{N} \right) = \frac{2.8}{640N + 450} \quad (4.1)$$

The collector current for the load transistor in saturation is

$$I_{C,sat} = \frac{3.6 - 0.2}{640} = 5.31 \text{ mA} \quad (4.2)$$

The value of N must satisfy the following relation,

$$h_{FE} \cdot I_B \geq I_{C,sat} \quad (4.3)$$

For $N = 5$, $I_B = 0.767 \text{ mA}$. Therefore, h_{FE} must be greater than 7.

4.3.3 Noise Margins

When the output is in 0 state $V_O = 0.2 \text{ V}$. If this voltage becomes about 0.5 V (cut-in voltage of transistor), the load transistor comes to conduction which causes malfunction of the circuit. Hence, the logic 0 noise margin $\Delta 0 \approx 0.3 \text{ V}$.

The logic 1 noise margin depends upon the number of gates being driven. For $N = 5$,

$$V_O = \frac{90}{90 + 640} \times (3.6) + \frac{640}{90 + 640} \times (0.8) = 1.14 \text{ V} \quad (4.4)$$

$$V_O = \frac{2.8 \times 90}{640 + 90} + 0.8$$

For $h_{FE} = 10$, the total base current required for load transistors to be driven into saturation will be $5 \times \left(\frac{5.31}{10} \right) \text{ mA}$ and the corresponding V_O must be 1.04 V . Therefore, the noise margin for 1 level is $\Delta 1 = 1.14 - 1.04 = 0.1 \text{ V}$.

2.

a) Draw the circuit of a 3-input TTL NAND gate and explain how it works.

Already noted.

b) Show the working principle and applications of CMOS inverter.

a. **Input HIGH (1):**

- i. The **NMOS** transistor turns on (allows current to flow), and the **PMOS** transistor turns off (stops current flow).
- ii. This pulls the output to LOW (0).

b. **Input LOW (0):**

- i. The **PMOS** transistor turns on (allows current to flow), and the **NMOS** transistor turns off (stops current flow).
- ii. This pulls the output to HIGH (1).

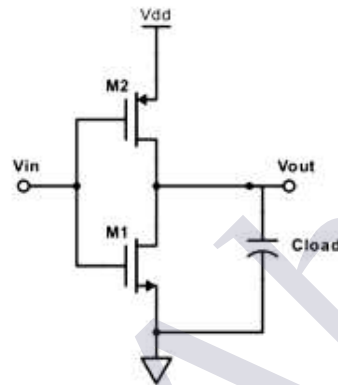


Figure : A static inverter with capacitive load

c) Discuss the speed of operation and fan out of logic gates.

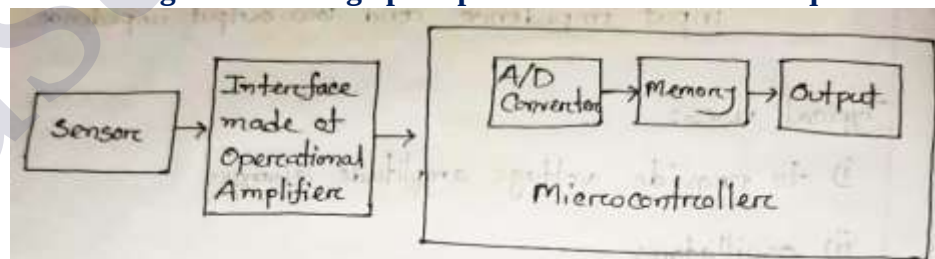
Already noted.

3.

a) Show the circuit symbol and terminals of operational amplifier 741C.

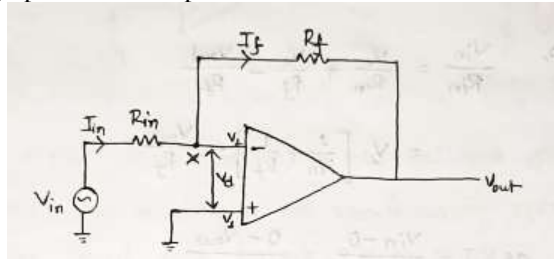
Already noted.

b) Show the block diagram for using op-amp in sensor-based data acquisition.



c) Draw the circuit for an inverting amplifier. Deduce the equation of output voltage, current and closed loop gain for inverting amplifier.

The circuit of the inverting operational amplifier is drawn below-

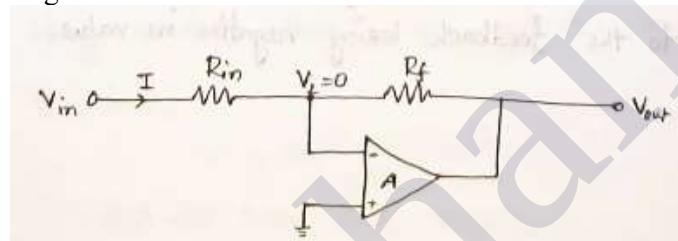


For an ideal amplifier, we know

- i) No current flows into the input terminals.
- ii) The differential input voltage is zero as $V_1 = V_2 = 0$ (virtual ground)

Then, by using these rules, we can derive the equations for calculating current, closed loop gain and output voltage of inverting amplifier.

Current 'I' flows through the resistor network as shown below-



So, Current, $I = \frac{V_{in} - V_{out}}{R_{in} + R_f}$

Therefore, $I = \frac{V_{in} - V_2}{R_{in}} = \frac{V_2 - V_{out}}{R_f}$

$\Rightarrow I = \frac{V_{in} - 0}{R_{in}} - \frac{V_2}{R_{in}} = \frac{V_2}{R_f} - \frac{V_{out}}{R_f}$

So, $\frac{V_{in}}{R_{in}} = \frac{V_2}{R_{in}} + \frac{V_2}{R_f} - \frac{V_{out}}{R_f}$

$= V_2 \left[\frac{1}{R_{in}} + \frac{1}{R_f} \right] - \frac{V_{out}}{R_f}$

And, as $I = \frac{V_{in} - 0}{R_{in}} = \frac{0 - V_{out}}{R_f}$

$\Rightarrow \frac{R_f}{R_{in}} = \frac{0 - V_{out}}{V_{in} - 0}$

The closed loop gain A_{cl} is given as, $\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$

and this can be transposed to give V_{out} as,

$V_{out} = -\frac{R_f}{R_{in}} \times V_{in}$

The negative sign is the indication of the inversion of the output signal w.r.t input signal as it is 180° out of phase. It is due to the feedback being negative in value.

4.

a) Explain how an operational amplifier can be used to design voltage follower?

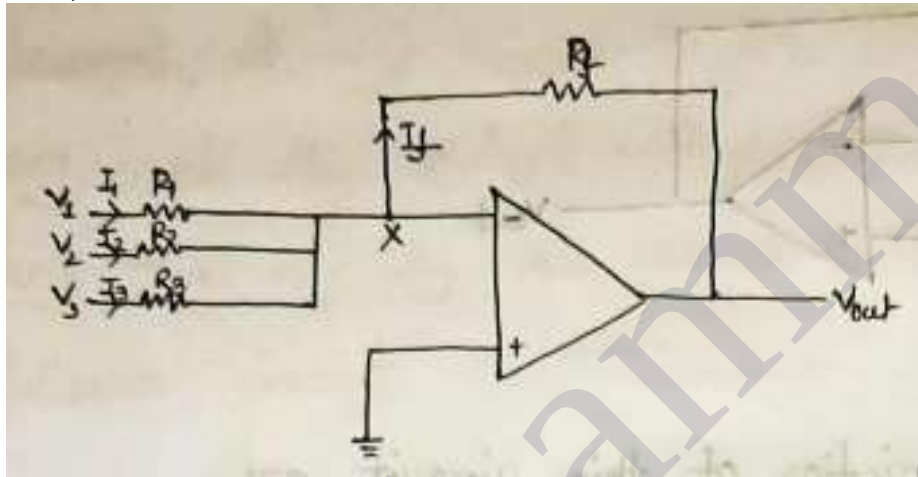
Already noted

b) Show the circuit diagram of integrator and differentiator and explain how it works.

Already noted.

c) How an operational amplifier can be used to build inverting adder?

Inverting adder,



We know, whenever there is a negative feedback to the op-amp then there is a virtual short between inverting and non-inverting terminal. So, if one terminal is at ground potential then another terminal also will be at the ground potential. Hence, the node 'x' will have 0 potential.

Now applying KCL at node 'x',

$$I_1 + I_2 + I_3 = I_f$$

$$\Rightarrow \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

Now let us assume $R_1 = R_2 = R_3 = R$, then

$$V_{out} = - \frac{R_f}{R} [V_1 + V_2 + V_3]$$

And if $R_f = R$, then

$$V_{out} = - [V_1 + V_2 + V_3].$$

In this way we can use op-amp as adder and we can add multiple input voltages.

5.

- a) Design a 4 bit unipolar D/A converter with $V(1)=-1$, $V(0)=0V$ and $R_F = 8R$. Find the analog output voltage of the D/A converter for each of the digital inputs from 0000 to 1111.

In the general formula (10.6) we put $N=4$ and find:

$$V_{out} = (b_3 \times 2^3 + b_2 \times 2^2 + b_1 \times 2^1 + b_0 \times 2^0)$$

At first put the combination 0000 in equation and the result is 0. Similarly, we can find for every 16 combination like the following table:

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

- b) Design a 3 bit parallel-comparator A/D converter and explain how it works.

Parallel comparator A/D converter: In a resistor voltage divider network, if you have N resistors of equal value R, the total resistance between V_{ref} and ground is NR, where N is the number of resistors.

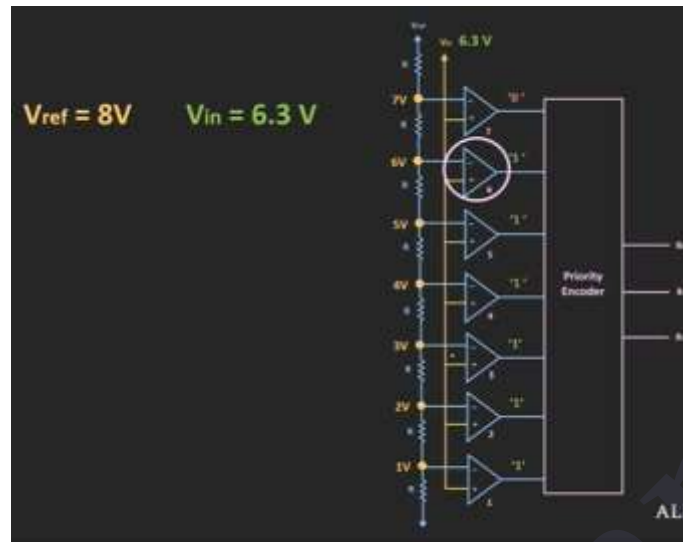
[Click here and look over the video and clear the concept.](#)

For the voltage at the n -th point (with n resistors below it):

$$V_n = V_{ref} \times \frac{nR}{NR}$$

This simplifies to:

$$V_n = V_{ref} \times \frac{n}{N}$$

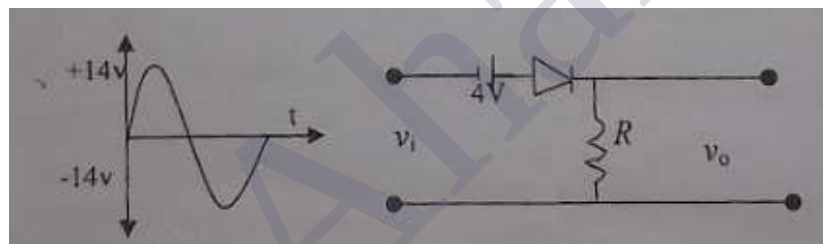


c) What is quantization error? How it is reduced in A/D converter?

Already noted.

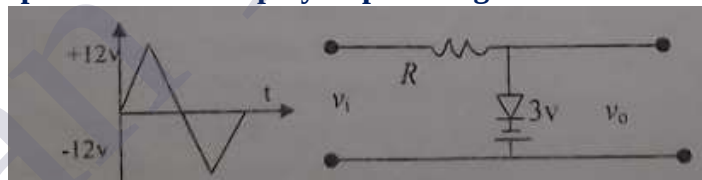
6.

a) Analyze the clipper network with given input and determine the output waveform of the given network.



Output voltage for positive half-cycle is +18 and for Negative half cycle +4.

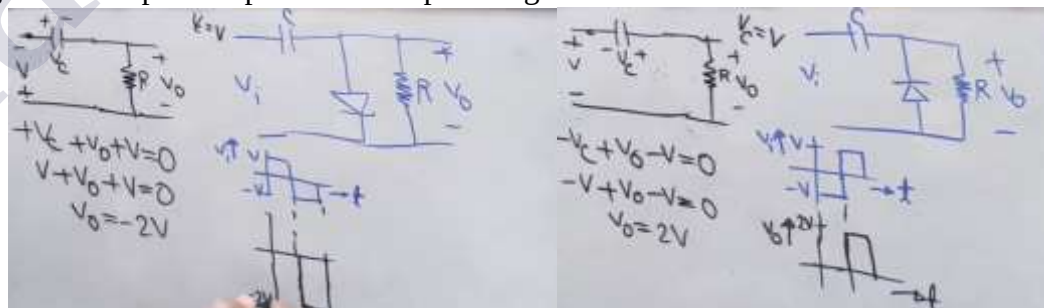
b) Determine the output waveform step by step of the given network.



For positive half-cycle the result is only -3 and for the negative half-cycle the result is -12 after a certain portion of the output; $V_i > 3$.

c) Design the clamping network and show the analysis of the network.

Negative Clamper and positive Clamper design in below:



1.

a) Discuss the bipolar logic families for electronics devices.

Bipolar logic families use bipolar transistors to perform digital logic operations.

- **Resistor-Transistor Logic (RTL)** is the oldest, featuring resistors and transistors; it is slow, consumes a lot of power, and is largely outdated.
- **Diode-Transistor Logic (DTL)** improves speed by using diodes along with transistors but still has moderate power consumption and complexity.
- **Transistor-Transistor Logic (TTL)**, the most common of the three, uses only transistors for both logic and amplification, offering faster operation but higher power consumption. TTL was widely used before CMOS technology became more prevalent due to its lower power usage and better integration

b) Draw the circuit diagram of 3-input DTL NAND Gate Calculate fan out, noise margin and average power if the parameters are considered as voltage across a conducting diode Cut in voltage of diode =0.6 V; Cut in voltage of transistor =0.5 V; $V_{BE,sat} = 0.8$; $V_{CE,sat} = 0.2$; $h_{FE} = 20$.

Same as 2018-2019; 2(a)

c) Show the limitations of DTL and TTL circuits.

➤ Limitations of DTL (Diode-Transistor Logic):

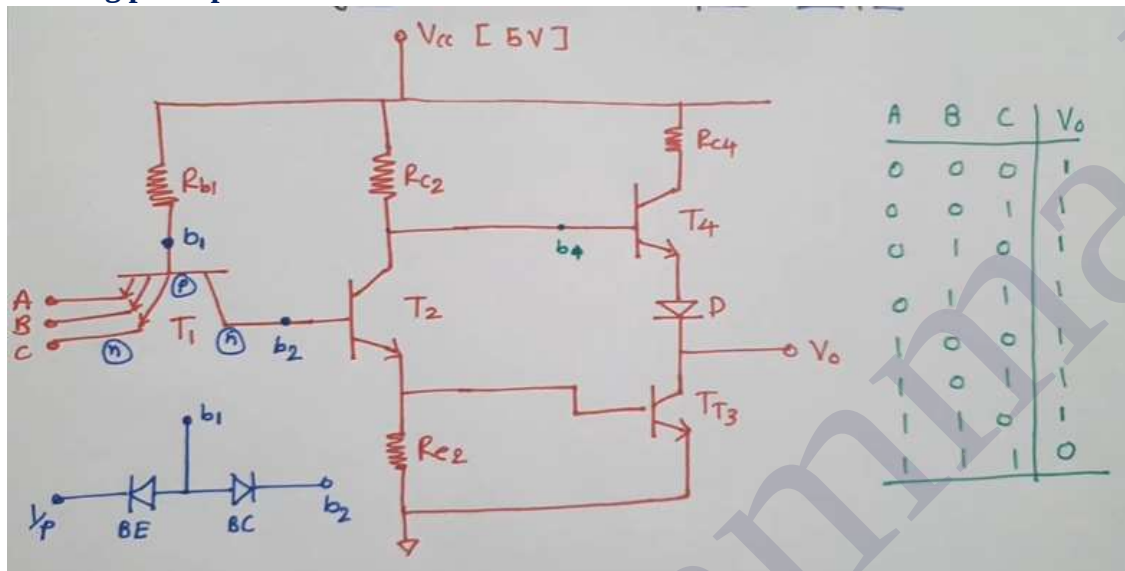
1. **Speed:** Slower than TTL because diodes add delays.
2. **Power Consumption:** Higher power consumption compared to TTL due to diode forward voltage drop.
3. **Complexity:** More complex design and less integration compared to modern logic families.

➤ Limitations of TTL (Transistor-Transistor Logic):

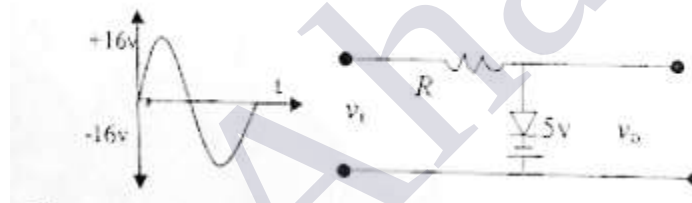
1. **Power Consumption:** Higher power consumption compared to CMOS due to continuous current flow.
2. **Speed:** While faster than RTL and DTL, it is slower than CMOS technology.
3. **Integration:** Limited integration compared to CMOS, which can affect the number of gates on a single chip.

2.

- a) Draw the circuit of a TTL gate with totem-pole output driver. Write down the working principle of the circuit.



- b) Determine the output waveform step by step of the given network.

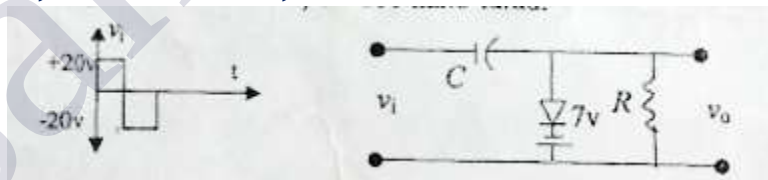


-5v for positive cycle, for negative half cycle $V_i = 5$ when the $V_i < 5$ and then $V_o = V_i$

- c) How figure of merit is measured for a digital IC?

3.

- a) Analyze output voltage and time constant is analyzed in given clamping network where $f = 2000$ Hz, $R = 20$ Kilo-Ohm, $C = 100$ nano-farad.



-7v output in positive supply and -47 output in negative supply.

- b) What are differences between open loop gain and close loop gain of operational amplifier?

Already noted.

- c) Show how can an operational amplifier be used as an adder and subtractor?

Already noted.

4.

a) Explain how an operational amplifier can be used as an integrator?

Already noted.

b) Show how a low pass filter is designed using the operational amplifier?

c) How voltage gain is obtained for inverting amplifier?

In an **inverting amplifier**, the voltage gain is determined by the ratio of two resistors:

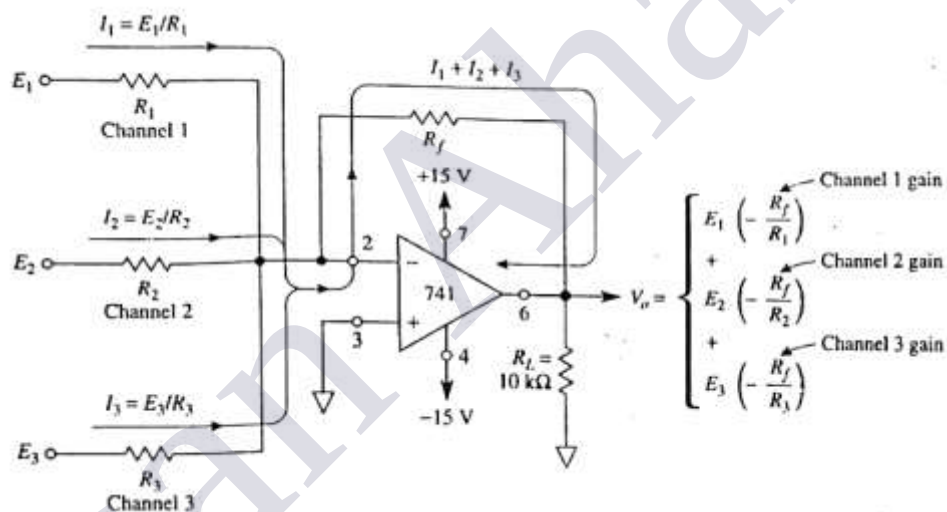
$$\text{Gain} = -R_f/R_{in}$$

- **R_f** is the resistor connected between the output and the inverting input.
- **R_{in}** is the resistor connected to the input signal.

The **negative sign** means the output is flipped or "inverted" compared to the input.

5.

a) Draw the circuit diagram of a multichannel amplifier. Show the output voltage analysis and circuit diagram of differentiator.

**Second part answer already noted.**

b) Explain how an op-amp can be used as voltage follower?

Already noted.

c) Give the examples where D/A and A/D converters are used.

Digital-to-Analog (D/A) and Analog-to-Digital (A/D) converters are used in many everyday devices. Here are some examples:

- **D/A Converter:** Used in **audio systems** to convert digital music (like from a smartphone) into analog signals that speakers can play as sound.
- **A/D Converter:** Used in **microphones** to convert analog sound waves into digital signals for recording or processing on a computer.

6.

- a) Deduce the output voltage for the weighted-resistor D/A converter. Why offset voltage is used 3+ for D/A converter?

The circuit of Fig. 10.1 can be used for converting the digital input to analog output which operates according to the above principle. This circuit is referred to as a *weighted-resistor D/A converter* since the resistance values are weighted in accordance with the binary weights.

In the circuit Fig. 10.1, the digital inputs (1 or 0) operate the switches. A switch is thrown to position 1 or 0 for a digital input corresponding to that bit being 1 or 0,

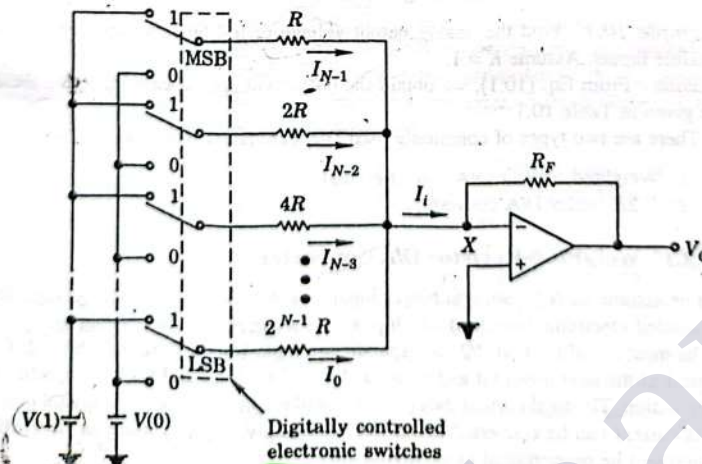


Fig. 10.1 Weighted-resistor D/A converter.

respectively. The voltage applied to a resistor is $V(1)$ if the switch connected to it is in position 1 and $V(0)$ if it is in position 0. The current, I_i is given by

$$I_i = I_{N-1} + I_{N-2} + I_{N-3} + \dots + I_2 + I_1 + I_0 \quad (10.2)$$

where

$$\left. \begin{aligned} I_{N-1} &= V_{N-1} / R \\ I_{N-2} &= V_{N-2} / 2R \\ I_{N-3} &= V_{N-3} / 2^2 R \\ &\vdots \\ I_0 &= V_0 / 2^{N-1} R \end{aligned} \right\} \text{where } V_n = \begin{cases} V(1) & \text{if } b_n = 1 \\ V(0) & \text{if } b_n = 0 \end{cases} \quad (10.3)$$

For straight binary input, $V(0) = 0$ and $V(1) = -V_R$, and the output voltage V_O is given by

$$V_O = -(-V_R) \left(\frac{R_F}{R} b_{N-1} + \frac{R_F}{2R} b_{N-2} + \frac{R_F}{2^2 R} b_{N-3} + \dots + \frac{R_F}{2^{N-1} R} b_0 \right) \quad (10.4)$$

which is of the same form as Eq. (10.1)

$$\text{with} \quad K = \frac{R_F}{2^{N-1} R} \cdot V_R \quad (10.5)$$

The output swings in only one direction and therefore is *unipolar*. If it is required to convert digital data in bipolar format such as in sign-magnitude, 1's complement or 2's complement format, then $V(0) \neq 0$. In such cases, $V(0)$ is used to *offset* the output swing.

With $V(1)$ and $V(0)$ as the voltages applied to the resistor network for 1 and 0, respectively, the output voltage V_O of Fig. 10.1 is given by

$$V_O = \frac{R_F}{2^{N-1} R} (2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + \dots + 2^1 V_1 + 2^0 V_0) \quad (10.6)$$

- b) Find the analog output voltage of a 5-bit D/A converter where $K=1$, Design the D/A converter for this.**

$$V_{\text{out}} = K \times (b_4 \times 2^4 + b_3 \times 2^3 + b_2 \times 2^2 + b_1 \times 2^1 + 0 \times 2^0)$$

Use this formula and draw truth table and explain the answer.

- c) How 4 bit digital output can be obtained from an analog voltage in the range of 0 to V?**

$$V_{\text{out}} = K \times (b_3 \times 2^3 + b_2 \times 2^2 + b_1 \times 2^1 + 0 \times 2^0)$$

1.

a) Write down the short note on i) Figure of Merit ii) Fan out iii) Noise immunity.

Already noted.

b) Draw the circuit diagram of 3-input DTL NAND Gate. Explain the operation of DTL NAND gate.

Already noted.

c) Show the limitations of RTL and DTL circuits.

➤ **Limitations of RTL (Resistor-Transistor Logic):**

1. **Speed:** RTL circuits are slower because resistors cause delays and limit current flow.
2. **Power Consumption:** High power consumption due to resistors constantly drawing current.
3. **Noise Margins:** Poor noise margins, making them less reliable in noisy environments.

➤ **Limitations of DTL (Diode-Transistor Logic):**

4. **Speed:** Slower than TTL because diodes add delays.
5. **Power Consumption:** Higher power consumption compared to TTL due to diode forward voltage drop.
6. **Complexity:** More complex design and less integration compared to modern logic families.

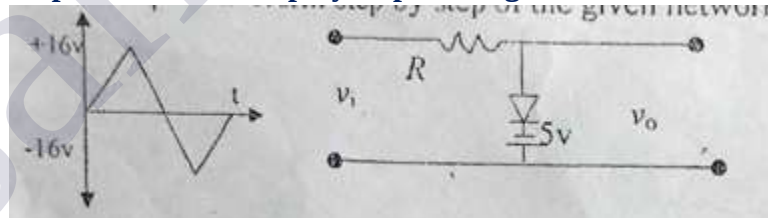
2.

a) Why clipper is used in electronic circuits.

A clipper is used in electronic circuits to prevent the voltage from going beyond a certain level. It "clips" the parts of the signal that exceed a set limit, protecting components from too much voltage.

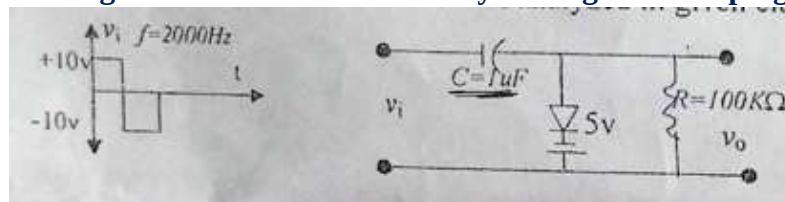
Example: In a radio, a clipper might be used to keep the signal within a safe range, ensuring the sound doesn't get distorted or damage the speaker.

b) Determine the output waveform step by step of the given network.



-5v for positive cycle and $V_i = 5$ when the $V_i < 5$ and then $V_o = V_i$

c) Analyze output voltage and time constant is analyzed in given clamping network?



-5v for positive supply and -25v for negative supply

3.

a) Write down the characteristics of ideal op-amp.

- i. Infinite input impedance. ($R_i = \infty$)
- ii. Output impedance. ($R_o = 0$)
- iii. Infinite open-loop gain. ($A = \infty$)
- iv. $V_{out} = 0$ when $V_{in} = 0$
- v. Infinite bandwidth.
- vi. Infinite Common mode rejection rate (CMRR).

b) What is open-loop voltage gain and closed-loop gain? Show the pin configuration of a typical op-amp.

Already noted

4.

a) How voltage gain is obtained for inverting amplifier?

Already noted.

5.

a) Explain why capacitor and inductor are very useful in electric circuits.

Capacitors and inductors are useful in electric circuits because they store and manage energy:

- **Capacitors** store energy in an **electric field** and can release it quickly. They are used to smooth out voltage, filter signals, or store charge temporarily, like in cameras or power supplies.
- **Inductors** store energy in a **magnetic field** and resist changes in current. They are useful for controlling current flow, filtering signals, and tuning circuits, like in radios or transformers.

b) Design an analog computer circuit to solve the differential equation

$$\frac{d^2 v_0}{dt^2} + 3 \frac{dv_0}{dt} + 2v_0 = 4 \cos 10t, t > 0 \text{ subject to } V_0(0) = 2, V_0'(0) = 0$$

6.

a) Analyze the output voltage and K factor of the weighted-resistor D/A converter.

Already noted.b) Consider a 4-bit D/A converter with $V(1) = -1V$, $V(0) = 0V$ and $R = 16R$. Obtain the analog output voltage for each digital input where $V_o = 0V$ for the digital input of 1000. Draw the circuit of D/A converter and adjust the offset voltage for the circuit.**Same as $R = 8R$**

c) What is quantization error? How quantization error can be reduced in analog to digital.

Already noted.